COMPUTER SCIENCE PRACTICAL FILE

CODE NO – 083 (2025-26)
ON PYTHON PROGRAMS

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Class : 12th A2
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Sr. No.	Program/Topic	Signature
	PYTHON REVISION TOUR	
01	Write a program to test that tuples are immutable.	
02	Write a program that INPUTS an integer in range 0-999 and then prints if the integer entered is a 1/2/3-digit number.	
03	Write a program to verify that the sum of first n odd natural numbers: $1+3+5+\ldots\ \ldots + (2^{n-1}) = n^2$	
04	Write a random number generator that generates random numbers between 1 and 6 (simulates a dice).	
05	Write a program to display the addition and subtraction of two matrices.	
	USER DEFINED FUNCTIIONS	
06	Write a program to calculate cube of given number through function.	
07	Write a program to demonstrates the functioning of default arguments.	
08	Write a program to reads a line and prints its statistics like: ➢ Number of uppercase letters. ➢ Number of lowercase letters. ➢ Number of alphabets. ➢ Number of digits	
09	Write a program with a user-defined function with string as a parameter to check whether passed string is a palindrome or not.	
10	Write a program with a user-defined function with string as a parameter which replaces all vowels in the string with '#'.	

FILE HANDLING (TEXT FILES)		
11	Write a program to read a text file and display the number of vowels/ consonants/uppercase/lowercase characters and other than character and digit in the file.	
12	Write a program to displaying the size of a file after removing EOL characters, leading and trailing white spaces and blank lines. (Use of strip())	
13	Write a Python code to find the size of the file in bytes, the number of lines, number of words and number of characters.	
14	Write a program to remove all the lines that contain the character `w' in a file and write it to another file.	
15	Write a program that copies a text file 'POEM1.TXT' onto 'NEWPOEM.TXT' barring the lines starting with a "@" sign.	
FILE HANDLING (CSV FILES)		
16	Write a program to perform read and write operation in a csv file (student.csv) having fields as roll number, name, stream and percentage.	
17	Write a program to search the record of a particular student from CSV file on the basis of inputted name.	
18	Write a program to copy data from one (STUDENT.csv) file to another (STUDENT1.csv) csv file. It should also display the contents of the file STUDENT1.csv.	
FILE HANDLING (BINARY FILES)		
19	Create a binary file with name and roll number. Search for a given roll number and display the name, if not found display appropriate message. 1. Inserting/Appending 2. Reading/Display 3. Searching.	

20	Create a binary file with roll number, name and marks. Input a roll number and update details. 1. Inserting/Appending 2. Reading/Display 3. Updating/Modify	
21	Write a menu driven interactive program to get employee records (empno, name, dept and salary) from the user and write into a binary file. The program should be able to get data from the user and write into the file as long as the user wants. It should also read and display all the records from 'EMP.DAT' file.	
22	Write a program to enter a student's details (roll number, name and marks) and write into a binary file student.dat using pickling. Write code to display the record as well as display a message whether the student is eligible for a merit certificate or not.	
	Using Python Libraries	
23	Create a module LENGTH_CONV.py that stores functions for various lengths conversion e.g., ➢ miletokm() to convert miles to kilometre ➢ kmtomile() to convert kilometres to miles ➢ feettoinches() ➢ inchestofeet() It should also store constant values such as value of (mile in kilometres and vice versa). [1 mile =1.609344 kilometre; 1 feet =12 inches] help() function should display proper information	
24	Write a program in Python that calculates the following: 1. Area of a circle. 2. Circumference of a circle. 3. Area of a rectangle. 4. Perimeter of a rectangle. 5. Exit. Create respective modules for each of the operations and call them separately using a menu-driven program.	
25	Write a program to create a library in python and import it in a program.	
26	Write a program to demonstrates the use of random module.	

Data Structure – Linear Lists

27	Write a python program to implement a stack using a list data-structure.	
28	Write a program that maintains students records (Rollno, Name, and Marks) depending upon user's choice, either pushes or pops records in the form of nested list in a stack.	
29	Alam has a list containing 10 integers. You need to help him create a program with separate user defined functions to perform the following operations based on this list. ➢ *Traverse the content of the list and push the even numbers into a stack.* ➢ *Pop and display the content of the stack.* For Example: If the sample Content of the list is as follows: N =[12, 13, 34, 56, 21, 79, 98, 22, 35, 38] Sample Output of the code should be: 38 22 98 56 34 12	
30	Write a python program to implement searching methods based on user choice using a list data-structure. (linear & binary)	
31	Program for creating and traversing a two-dimensional list i.e., 2D.	
32	Program for inserting an element in a sorted array using bisect module	

Database Query using SQL

33	Create a student table and insert data. Implement the following SQL commands on the student table:	

Name of column	Data Type
Adm_no	Number(4)
Name	Varchar(30) Primary Key
Stipend	Number(9, 2)
Stream	Varchar(30)
AvgMarks	Number(3,2)
Grade	Char(1)
Class	Char(3)

a) Write command to create above table with suitable constraints. (Adm_no as PRIMARY KEY)
b) Select/display all the Nonmedical stream students from STUDENT.
c) List the names of those students who are in class 12 sorted by Stipend.
d) List all students sorted Avg in descending order.
e) Display a report, listing Name, Stipend, Stream, and amount of stipend received in a year assuming that the Stipend is paid every month.

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Create a table EMP and insert data. Implement the following SQL commands on the student table:

Name of column	Data Type
EMPNO	Number(4)
ENAME	Varchar(30) Primary Key
JOB	Varchar(30)
MGR	Number(04)
HIREDATE	Date
SAL	Number(2)
COMM	Number(9, 2)
DEPTNO	Number(9, 2)

a) Write command to create above table.
b) Display all the records (all columns) from table EMP who are not getting commission.
c) Write a query to display employee number, name, SAL and SAL*12 as annual salary whose commission is not NULL from table EMP.
d) List the details of all employees whose annual salary between 25000 – 40000.
e) To count the number of employees in EMP table.

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Harshit, who works as a database designer, has developed a database for an Airline. This database includes a table FLIGHT whose column (attribute) names are mentioned below.

FLCode: Shows the Flight code

Start: Shows the starting place

Destination: Shows the destination place

No_of_Stops: Indicates the number of stops

No_of_Flights: Indicates the number of flights

Write SQL query to:

i) Write command to create above table.

ii) Display the flight codes, starting place, destination, number of flights in ascending order of number of flights.

	iii) Display flight codes and starting place whose destination is Jaipur. iv) Display flight codes, starting place and destination name whose number of flights is 0 v) Display the starting place and destination name whose number of stops is 0.	

Table: FLIGHT

FLCode	Start	Destination	No_of_Stops	No_of_Flights
IC101	Delhi	Agartala	1	5
IC102	Mumbai	Sikkim	1	3
IC103	Delhi	Jaipur	0	7
IC105	Kanpur	Chennai	2	2
IC107	Mumbai	Kanpur	0	4
IC431	Indore	Chennai	3	2
IC121	Delhi	Ahmedabad	2	6